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A Comparative Analysis for Split Ring Resonators of Different Geometrical Shapes

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Abstract—A comparative study of split ring resonator (SRR) of different geometrical shape is presented in this paper. Closed form expressions for computing the resonance frequency of the circular, square and hexagonal shaped SRRs are also presented and compared. The surface current density of the SRRs is also studied using electromagnetic simulator and their comparative analysis is presented.

Keywords—Split Ring Resonators; Double negative media

I. INTRODUCTION

Double negative (DNG) media or negative index materials (NIM) can be realized using periodic alignment of structures exhibiting negative permittivity and negative permeability. Pendry showed that while negative permittivity is realized using conducting wires, negative permeability can be realized using split ring resonators (SRR) [1]. Different researchers have worked with different configuration of SRR viz. circular SRR (C-SRR), square SRR (S-SRR), hexagonal SRR (H-SRR) etc. for realizing artificial magnetic material or employing the SRR as perturbations for different passive planar circuit design. Circular and square split ring resonators have been extensively studied by [1]-[7]. C-SRR, S-SRR and H-SRR has found lots of applications in passive planar circuit design for last few years like filter, power divider, duplexers and phase shifter [8]-[12]. In this paper, a comparative study of SRRs having different geometrical configurations is presented. The resonance frequencies of the circular, square and hexagonal SRRs are computed using the simplified theoretical models and compared with results obtained from a commercial EM solver [13]. All three geometrical configurations of the SRR, printed on similar dielectric substrates, are studied for same physical dimensions. The comparison of their resonance frequency, which determines the region of negative permeability, suggests that for similar physical dimension the H-SRR resonates at higher frequency. A comparative study of surface current density of the SRRs obtained using EM simulator is also studied and presented.

II. THEORY

Fig. 1 shows a schematic view of a C-SRR, S-SRR and H-SRR having strip width c and spacing d between the

rings. g_1 and g_2 are gaps within the inner ring and outer ring, respectively. The structure is printed on a dielectric substrate with dielectric constant, ϵ_r and thickness h . The equivalent circuit model of these split ring resonators is shown in Fig.2. Under the exposure of external magnetic field the induced electromotive force around the SRR causes a currents which passes from one ring to the other through the inter ring spacing, d and the structure behaves as an oscillatory L-C circuit.

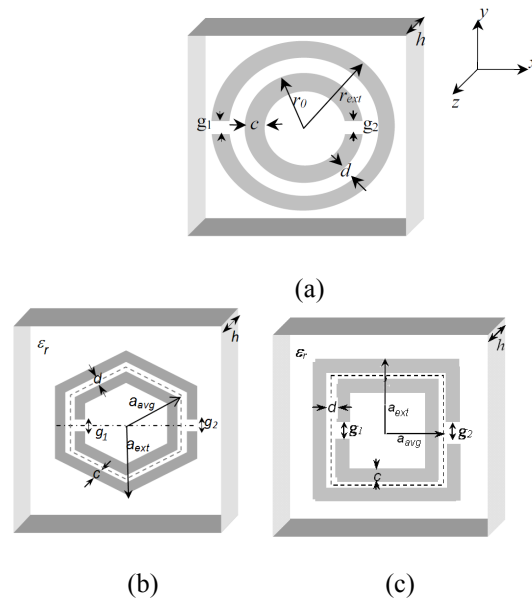


Figure 1. Schematic view of split ring resonators formed with metallic strips of width, c , dimension r_{ext} (For Circular) and a_{ext} (For Square and Hexagonal), with inter ring spacing d and split gap dimensions, $g_1 = g_2$, printed on a dielectric substrate having thickness, h and dielectric constant ϵ_r . (a) Circular (b) square (c) Hexagonal

The resonance frequency of the SRR is given by,

$$\omega_0 = \sqrt{\frac{1}{L_T C_{eq}}} \quad (1)$$

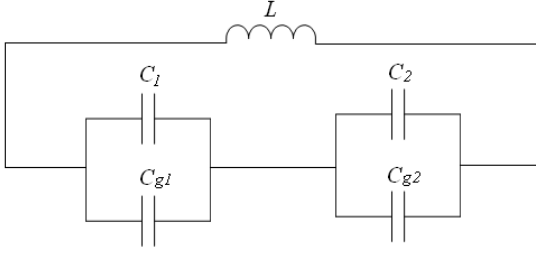


Figure 2. Equivalent circuit of the Split Ring Resonators shown in Fig. 1

The expression of the resonant frequency of different SRR is given as

For C - SRR :

$$f_{0C} = \frac{1}{2\pi\sqrt{LC_{eq}}} = \frac{1}{2\pi\sqrt{L_T \left[\frac{(\pi r_0 - g)C_{pul}}{2} + \frac{\epsilon_0 ct}{2g} \right]}} \quad (2)$$

For S - SRR :

$$f_{0S} = \frac{1}{2\pi\sqrt{L_T C_{eq}}} = \frac{1}{2\pi\sqrt{L_T \left[\left(2a_{avg} - \frac{g}{2} \right) C_{pul} + \frac{\epsilon_0 ch}{2g} \right]}} \quad (3)$$

For H - SRR :

$$f_{0H} = \frac{1}{2\pi\sqrt{L_T C_{eq}}} = \frac{1}{2\pi\sqrt{L_T \left[\frac{(3a_{avg} - g_1)C_{pul}}{2} + \frac{\epsilon_0 ch}{2g_1} \right]}} \quad (4)$$

where L_T is the total inductance of the SRR and C_{eq} is the equivalent capacitance of the structure and is calculated from the equivalent circuit in Fig.2, as in [5].

C_{pul} , is the per unit length capacitance, between the rings and r_0 (C-SRR) and a_{avg} (S-SRR and H-SRR) are the distance of the two constituent rings of the SRRs from the centre. C_{pul} is calculated as [14],[5]

$$C_{pul} = \frac{\sqrt{\epsilon_e}}{c_0 Z_0} \quad (5)$$

where $c_0 = 3 \times 10^8$ m/s is the velocity of light in free space. ϵ_e is the effective permittivity of the medium and Z_0 is the impedance of the medium.

The total inductance of the structures, L_T , is computed as [15]

$$L_T = 0.00508l \left(2.303 \log_{10} \frac{4l}{d} - \theta \right) \quad (6)$$

Table 1. Comparison of computed and simulated resonance frequencies of square, circular and hexagonal shaped SRR for different r_{ext} and C values. Parameters: $d=0.2$ mm, $g=0.2$ mm, $\epsilon_r = 2.33$, $h=1.575$ mm.

SRR dimension (mm)		SRR Resonance frequency (GHz)					
		f_{oS}		f_{oC}		f_{oH}	
r_{ext} / a_{ext}	c	Computed	Simulated	Computed	Simulated	Computed	Simulated
2.2	0.4	5.85	5.66	7.24	7.12	7.97	7.54
	0.5	6.16	6.24	7.60	7.59	8.42	8.18
	0.6	6.51	6.73	8.01	7.89	8.42	8.69
2.4	0.4	5.19	5.03	6.42	6.23	7.07	6.99
	0.5	5.44	5.51	6.71	6.6	7.42	7.18
	0.6	5.71	5.99	7.04	7.07	7.81	7.64
2.6	0.4	4.65	4.54	5.77	5.65	6.33	5.97
	0.5	4.86	4.85	6.00	5.95	6.62	6.34
	0.6	5.08	5.12	6.26	6.17	6.94	6.76

where, l and d are the wire length and width, respectively. The constant θ varies with wire geometry and is given as [15]

$$\begin{aligned} \theta &= 2.451 \text{ for C-SRR} \\ &= 2.853 \text{ for S-SRR} \\ &= 2.636 \text{ for H-SRR} \end{aligned} \quad (7)$$

III. RESULTS

The resonance frequency for SRR of different geometries is computed using equations (2), (3) and (4). The SRR are printed on dielectric substrates having $\epsilon_r = 2.33$ and thickness, $h = 1.575$ mm. The simulated resonance frequency was extracted from the transmission coefficient curves when excited by EM wave as proposed in [4]. The computed values calculated using eqn. (2)- (4) are compared with the simulated values and are presented in Table I. The values were extracted for r_{ext} (C-SRR) and a_{ext} (for S-SRR and H-SRR), 2.2 mm, 2.4mm and 2.6mm. For each of these r_{ext} or a_{ext} , three different values of ring widths of the SRR $c = 0.4$ mm, 0.5mm and 0.6mm are considered. The tabulated comparison shows that for similar footprint with identical r_{ext} or a_{ext} values, the H-SRR resonates at around 36% of higher side of the spectrum compared to the S-SRR. The C-SRR resonates at higher frequency compared to the S-SRR and at lower frequency compared to the H-SRR, having similar dimensional parameters. The simulated S_{21} of the SRRs for different gap dimensions, $g=0.2$ mm and $g=0.8$ mm, are shown in Fig 3. As revealed from Table I, Fig. 3 depicts the shift in the resonance frequency with H-

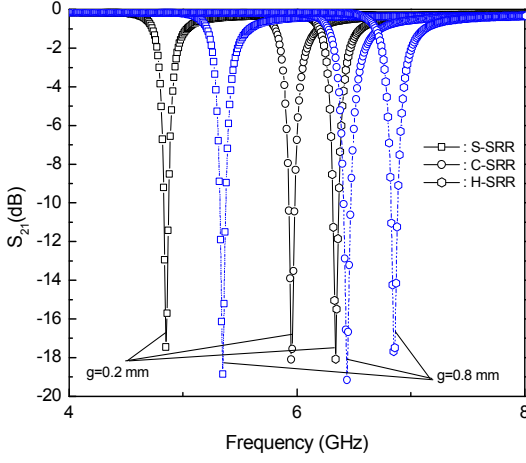


Fig.3. Simulated S_{21} of the C-SRR, S-SRR and H-SRR shown in Fig.1. $\epsilon_r = 2.33$, $c = 0.5\text{mm}$, $d = 0.2\text{mm}$, $h = 1.575\text{mm}$, $g_1 = g_2 = 0.2\text{mm}$, $a_{ext} = r_{ext} = 2.6\text{mm}$

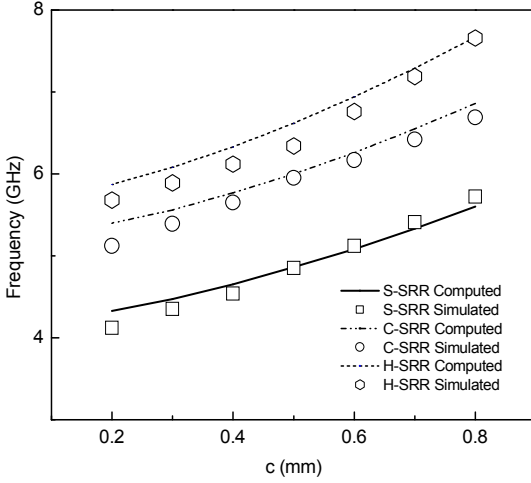


Fig. 4. Computed and Simulated resonance frequency of S-SRR, C-SRR and H-SRR as a function of c . $\epsilon_r = 2.33$, $d = 0.2\text{mm}$, $h = 1.575\text{mm}$, $g_1 = g_2 = 0.2\text{mm}$, $a_{ext} = r_{ext} = 2.6\text{mm}$.

SRR resonating at higher frequency compared to S-SRR and C-SRR.

Fig. 4 shows a plot of the computed and simulated resonance frequency as a function of the ring width c for all three SRR geometries. Close correspondence of the computed values when compared with the simulated data is revealed from the plot. Figure 5 shows the induced surface current density of the SRRs when excited with an incident EM signal. The figure shows higher current distribution in S-SRR which yields higher L_T and hence lower resonance frequency.

IV. CONCLUSIONS

A comparative study of SRR of different geometrical shapes is done using simple equivalent circuit approach. The

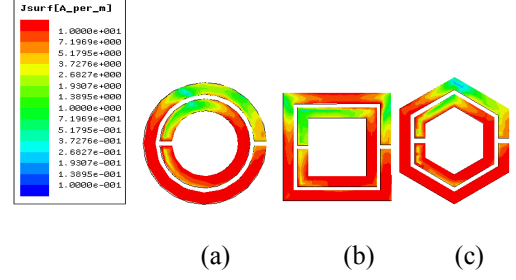


Fig. 5. Simulated surface current distributions of (a)Circular (b) square (c) Hexagonal shaped Split Ring Resonators

proposed comparative analysis is validated using commercial EM solver tool. The comparative analysis shows a frequency shift of around 36% for different geometrical shaped SRRs with similar footprints.

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